Maths Explore

Course notes

Air and antiderivative

For f continued and positive on $[a, b]$, $a \leq b$, integral $\int_a^b f(x)\ \mathrm{d}x$ is the area under the curve, above of the axis between $x = a$ and $x = b$, expressed in d ·air units.

Fundamental Theorem

If f is continuous over an interval I and $a \in I$, then $F_a(x) = \int_a^x f(t)\ \mathrm{d}t$ is the antiderivative of f who annule in a. For any antiderivative F,

$$\int_a^b f(x)\ \mathrm{d}x = F(b) - F(a).$$

Any continuous function over an interval admits antiderivatives. An integral is a number; a antiderivative is a function.

Demonstration in the increasing positive case. For $h > 0$, the rectangles give $hf(x) \leq F_a(x+h) - F_a(x) \leq hf(x+h)$. After division by h the rate of increase tends to $f(x)$. For $h < 0$, growth $f(x+h) \leq f(t) \leq f(x)$ on $[x+h, x]$. By integrating,

$$(-h)f(x+h) \leq F_a(x) - F_a(x+h) \leq (-h)f(x).$$

By dividing by $-h > 0$, we get

$$f(x+h) \leq \frac{F_a(x+h) - F_a(x)}{h} \leq f(x).$$

The two bounds tend to $f(x)$. Out of continuity, $F_a' = f$. Any other antiderivative is $F_a + C$, where the calculation formula at the bounds.

Properties

For continuous functions and bounds within the same range:

$$\int_a^a f = 0, \quad \int_b^a f = -\int_a^b f, \quad \int_a^c f = \int_a^b f + \int_b^c f, \quad \int_a^b (\lambda f + \mu g) = \lambda \int_a^b f + \mu \int_a^b g.$$

If $a \leq b$ and $f \leq g$, then $\int_a^b f \leq \int_a^b g$. In particular, if $m \leq f \leq M$, then $m(b-a) \leq \int_a^b f \leq M(b-a)$. For any sign function, the integral is defined by antiderivatives.
between the curve and the east axis $\int_a^b |f|$, obtained by cutting off the sign changes. Between two curves, it's worth $\int_a^b |f - g|$.

Example. $\int_{-1}^1 x \, \mathrm{d}x = 0$, but the area between the right and the axis over this interval is 1.

Average value and integration by parts

For $a < b$, the average value is $\mu = \frac{1}{b-a} \int_a^b f(x) \, \mathrm{d}x$. The height rectangle μ and width $b - a$ has the same algebraic area.

Integration by parties

If u, v are of class $\mathcal{C}^1$ on $[a, b]$,

$$\int_a^b u(x)v'(x) \, \mathrm{d}x = [u(x)v(x)]_a^b - \int_a^b u'(x)v(x) \, \mathrm{d}x.$$

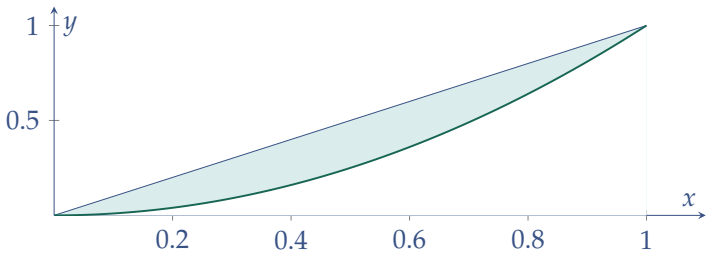
Demonstration. Integrate $(uv)' = u'v + uv'$ then isolate the desired integral. The hook is indispensable.

Example. With $u = x$ and $v' = e^x$, $\int_0^1 xe^x \, \mathrm{d}x = [xe^x]_0^1 - \int_0^1 e^x \, \mathrm{d}x = e - (e-1) = 1$.

Rectangles and integrated sequences

For f continued and increasing on $[a, b]$, $a < b$ and $n \geq 1$ and $h = \frac{b-a}{n}$, amounts $h \sum_{k=0}^{n-1} f(a+kh)$ and $h \sum_{k=1}^{n} f(a+kh)$ The difference between them is $h(f(b) - f(a))$. For $I_n = \int_0^1 x^n f(x) \, \mathrm{d}x$, start with sign and comparisons of integrand. If $0 \leq f \leq M$, then $0 \leq I_n \leq \frac{M}{n+1}$; the gendarmes give $I_n \to 0$. A party-based integration may also provide a relationship between I_n and I_{n+1}.

Convention at zero level. In $I_n = \int_0^1 x^n f(x)\,\mathrm{d}x$, the expression x^0 means constant function 1, y included in bound 0. $I_0 = \int_0^1 f(x)\,\mathrm{d}x$; no undetermined calculation is necessary.



Between $y = x$ and $y = x^2$ on $[0,1]$ value $\frac{1}{6}$.

Linking area, antiderivative and approximation

8.1 Two areas, one integral

First use: S59 and S61.

Example Solved · A Change of Sign

On $[0;3]$, $f(x) = x - 1$ is negative before 1, positive after. $[x^2/2 - x]_0^3 = 3/2$. Geometrical area is $-\int_0^1 f + \int_1^3 f = 1/2 + 2 = 5/2$. Integral compensation between two signed contributions.

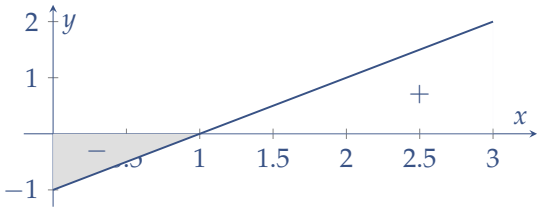


Figure 8.1. For a geometric area, the two regions add up; for integral, their signs Come in.

A1 · Immediate application For $g(x) = 2x - 2$ on $[0;3]$, calculate the integral and the area between the curve What is the relationship with the values of the example?

8.2 Why l ·aire gives a antiderivative

First use: S61.

Ownership · Fundamental Theorem

For f continues over interval I and $a \in I$, $A(x) = \int_a^x f(t)\,\mathrm{d}t$ is a antiderivative f. So, for any antiderivative F, $\int_a^b f = F(b) - F(a)$. The general case is admitted; the positive case increasing is shown below.

Demonstration. Suppose f continues, positive and increasing. For h>0, comparison to two rectangles given $f(x) \le [A(x+h) - A(x)]/h \le f(x+h)$. For h<0, it gives $f(x+h) \le [A(x+h) - A(x)]/h \le f(x)$. By continuity and bounds, the rate of increase tends to f(x); so $A' = f$. Any other antiderivative F differs from A of a constant; like A(a)=0, $A(b) = F(b) - F(a)$.□

Solved Example · Find a Derivative

For $A(x) = \int_0^x 2t\,\mathrm{d}t$, the area is x^2 when x is positive. The formula by antiderivative also gives $A(x) = x^2$ for x negative, with integral orientation. In both cases $A'(x) = 2x$.

A2 · Partial guidance On $[0;+\infty[$, pose $A(x) = \int_0^x e^{-t}\,\mathrm{d}t$. Give A(x), A(x), direction of variation and Interpret the boundary as an area.

8.3. Equivalent height, with its units

First use: S63.

Definition · Average value

For $a < b$, the average of $[a;b]$ is $m = \frac{1}{b-a}\int_a^b f$. If f is positive, the width rectangle b-a and height m has the same area as the area under the curve.

Example Solved · Flow and Volume

A flow rate $d(t) = 2 + t$ L/min for three minutes provides $\int_0^3 (2+t)\,\mathrm{d}t = 10{,}5$ L. Its flow rate Average is 10.5/3 $= 3.5$ L/min. The volume and average flow do not have the same unit.

A3 · Immediate application Calculate the average of x^2 on $[0;1]$. Trace the equivalent rectangle. Explain why this mean is not the average of the two values at the ends.

8.4 Draw and control the frame

First use: S63 and S66.

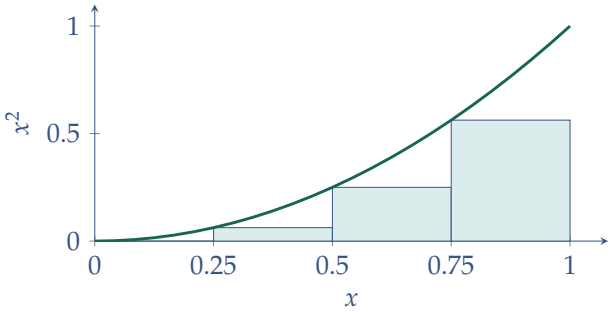


Figure 8.2. Rectangles to the left. Complete the drawing with the rectangles to the right and the right at height 1/3.

Example resolved · Two intervals

For x^2 on [0;1], two rectangles to the left give $L_2 = \frac{1}{2}(0^2 + (1/2)^2) = 1/8$; those on the right give $U_2 = \frac{1}{2}((1/2)^2 + 1^2) = 5/8$. The integral 1/3 is well between these bounds.

A4 · Search For $f(x) = x^2$ on $[0;1]$, cut in four equal intervals. Calculate amounts to the left and to the right; justify their positions in relation to the integral. intervals, their deviation is 1/n.

A first step: In the difference between the two amounts, the internal values are simplified.

8.5 Choosing what's drifting

First use: S66.

Property · Integration by parties

For class u,v $\mathcal{C}^1$ on $[a;b]$, $\int_a^b u'v = [uv]_a^b - \int_a^b uv'$. The formula results from the integration of $(uv)' = u'v + uv'$.

Example resolved · Two successive PPIs

For $I = \int_0^1 x^2 e^x \,\mathrm{d}x$, drifting the polynomial decreases its degree. First $I = [x^2e^x]_0^1 - 2\int_0^1 xe^x \,\mathrm{d}x$. Next $\int_0^1 xe^x \,\mathrm{d}x = [xe^x]_0^1 - \int_0^1 e^x \,\mathrm{d}x = e - (e-1) = 1$. So I=e-2. Write the bounds to each step avoids forgetting a constant.

A5 · Choice of strategy Calculate $\int_1^e \ln x \,\mathrm{d}x$. Justify your choice in IPP and check the result in derived from a antiderivative.

8.6 Frame before looking for a formula

First use: S66.

Solved Example · A Framework Without Explicit Primitive

Poisons $J_n = \int_0^1 x^n/(1+x)\,\mathrm{d}x$. Like 1 $/2 \le 1/(1+x) \le 1$, $1/[2(n+1)] \le J_n \le 1/(n+1)$. The two bounds tend towards 0, so $J_n \to 0$, without calculating Jn for each n.

A6 · Summary For $n \in \mathbb{N}$, $I_n = \int_0^1 x^n e^{-x} \,\mathrm{d}x$. Show that (I_n) is positive and decreasing. $e^{-1}/(n+1) \le I_n \le 1/(n+1)$, then its limit. Does an IPP give a relationship between I_n and I_{n+1} ?

A7 · Refuter Give a non-zero continuous function on $[-1;1]$ of which the integral is null. In addition, with continuity, it would be possible to conclude that an integral function Is zero identically zero?

Wait a minute at the courthouse. Explain on the figure why "integer null" and "area null" do not say Not the same thing.

Review postponed · S79. Calculate an area between trigonometric curves by keeping the search intersections and sign.

Research activity

Frame an integral with rectangles · 35 min.

We're trying to frame $I = \int_0^1 x^2 \,\mathrm{d}x$. For an integer $n \ge 1$, we share $[0;1]$ in n same intervals length. L_n the sum of the areas of rectangles whose heights are the values of x^2 to left ends, and U_n the one obtained at the straight ends.

1. For $n = 4$, represent the two families of rectangles, then calculate L_4 and U_4.

2. Justify that $L_4 \le I \le U_4$.

3. Write a Python argument function n which returns L_n and U_n. Algebraically determine $U_n - L_n$.
4. Determine the smallest integer n for which this framework has a width of not more than 0.01.
5. Calculate the exact value of I with the help of a antiderivative and verify that it belongs to the framework obtained for $n = 4$.